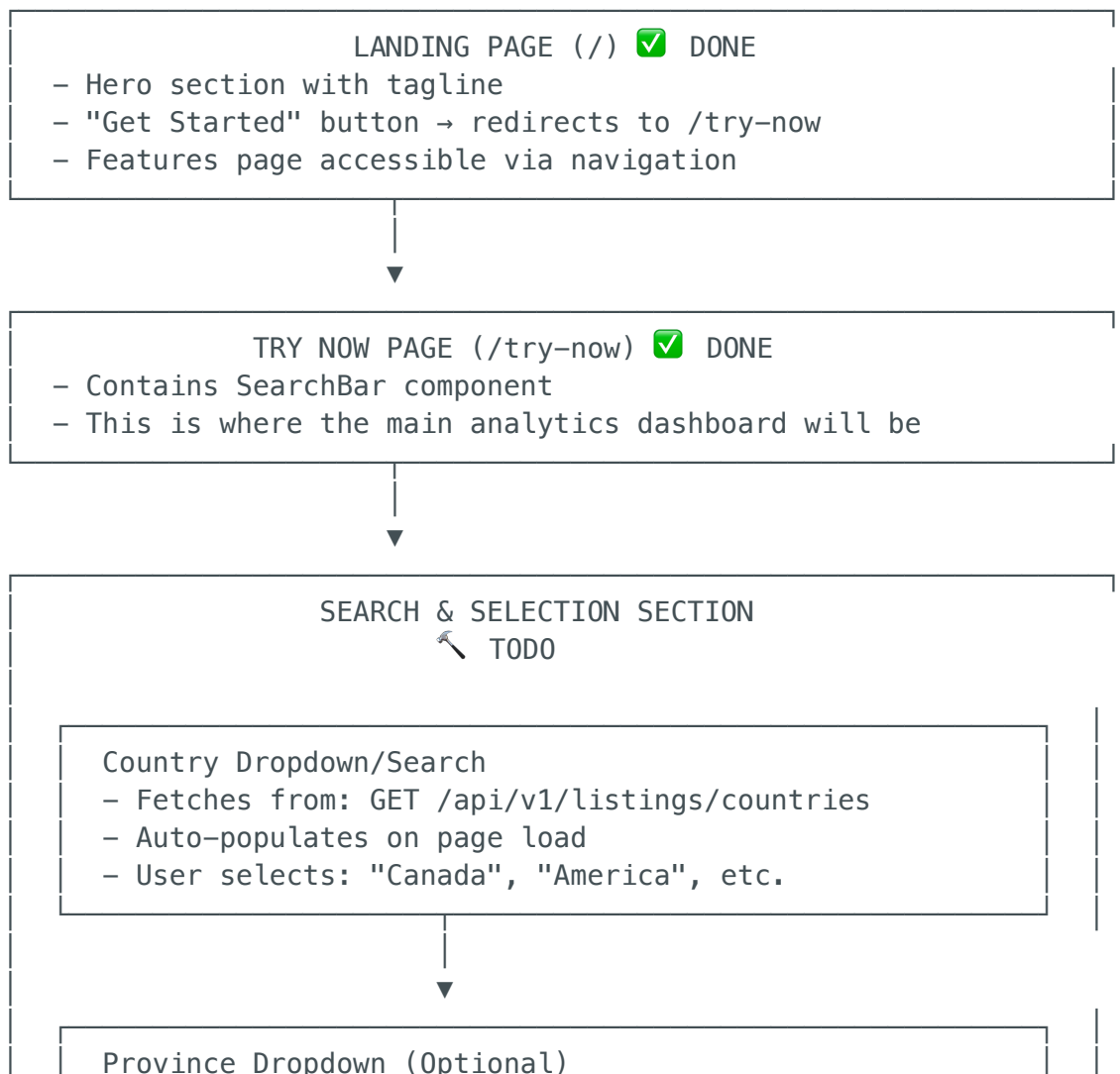


- Frontend Flowchart - Airbnb Analytics Project
 - 📊 Complete User Flow
 - 🏗️ Component Structure
 - 🔄 API Integration Flow
 - 📋 Implementation Checklist
 - ✅ Already Done
 - 🛠️ To Implement
 - 💡 Future Enhancements
 - 🎨 Design Considerations

Frontend Flowchart - Airbnb Analytics Project



Complete User Flow



- Fetches from: GET /api/v1/listings/provinces?country=
- Auto-populates based on selected country
- User can select: "Montreal", "Toronto", etc.
- If NOT selected: aggregates entire country



- "Show Analytics" Button
- Triggers API calls for analytics & listings



MAIN DASHBOARD LAYOUT (75% Analytics)

🔧 TODO

ANALYTICS SECTION (Left/Top 75%)

- Fetches: GET /api/v1/analytics/?country=X&province=Y

Chart 1: Property Type Distribution

- Material UI Pie/Bar Chart
- Data: property_type_distribution

Chart 2: Room Type Distribution

- Material UI Pie/Bar Chart
- Data: room_type_distribution

Chart 3: Top Amenities

- Material UI Bar Chart
- Data: amenity_distribution (top 20)

Price Statistics Cards

- Average, Highest, Lowest, Median
- Data: price_statistics

Review Statistics Cards

- Total, Average, Max, Min
- Data: review_statistics

Instant Booking Stats

- Pie Chart: Enabled vs Disabled
- Data: instant_booking_stats

LISTINGS SECTION (Right/Bottom 25%)

- Fetches: GET /api/v1/listings/?country=X&province=Y
- Scrollable list/table

Display columns per listing:

- id, name, listing_url
- picture_url (thumbnail)
- price, property_types, room_type
- number_of_reviews, bedrooms, bathrooms
- host_name, host_identity_verified

Features:

- Pagination (if > 100 listings)
- Click to view details (future: modal)
- Filter/sort options (future enhancement)



FUTURE ENHANCEMENTS



Listing Detail Modal/Page

- Full listing details with all fields
- Map view with latitude/longitude
- Amenities list



Advanced Filtering

- Price range slider
- Bedroom/bathroom filters
- Property type filter
- Instant booking filter



Comparison Mode

- Compare analytics between provinces
- Compare analytics between countries



Export Functionality

- Export listings to CSV
- Export analytics charts as images



User Preferences

- Save favorite locations
- Default country/province selection



Real-time Updates

- Auto-refresh data at intervals
- WebSocket support (if CSV files update)



Search/Filter within Listings

- Client-side filtering on loaded listings
- Search by name, host, property type



Component Structure

```
frontend/src/
├── pages/
│   ├── LandingPage/ ✓ DONE
│   │   └── LandingPage.tsx
│   ├── Features/ ✓ DONE
│   │   └── Features.tsx
│   ├── TryNow/ ✓ DONE (but needs implementation)
│   │   └── TryNow.tsx
│   └── AnalyticsDashboard/ ✎ TODO (NEW)
│       ├── AnalyticsDashboard.tsx
│       ├── AnalyticsDashboard.styles.ts
│       └── components/
│           ├── CountryProvinceSelector.tsx
│           └── AnalyticsCharts/
│               ├── PropertyTypeChart.tsx
│               ├── RoomTypeChart.tsx
│               ├── AmenityChart.tsx
│               ├── PriceStatsCard.tsx
│               ├── ReviewStatsCard.tsx
│               └── InstantBookingChart.tsx
│           └── ListingsList/
│               ├── ListingsList.tsx
│               ├── ListingCard.tsx
│               └── ListingCard.styles.ts
├── components/
│   ├── Header/ ✓ DONE
│   ├── Footer/ ✓ DONE
│   ├── Main/ ✓ DONE
│   └── Searchbar/ ✓ DONE (may need enhancement)
├── services/ ✎ TODO (NEW)
│   └── api.ts
│       ├── getCountries()
│       ├── getProvinces(country)
│       ├── getListings(country, province)
│       └── getAnalytics(country, province)
├── hooks/ ✎ TODO (NEW)
│   ├── useCountries.ts
│   ├── useProvinces.ts
│   ├── useListings.ts
│   └── useAnalytics.ts
└── utils/ ✎ TODO (NEW)
    ├── constants.ts
    └── API_BASE_URL
```



API Integration Flow

User Action Flow:

1. Page Load
 - ↳ GET /api/v1/listings/countries
 - ↳ Populate Country Dropdown
2. User Selects Country
 - ↳ GET /api/v1/listings/provinces?country=X
 - ↳ Populate Province Dropdown
3. User Clicks "Show Analytics"
 - ↳ GET /api/v1/analytics/?country=X&province=Y
 - ↳ Update Analytics Charts (75% of screen)
 - ↳ GET /api/v1/listings/?country=X&province=Y
 - ↳ Update Listings List (25% of screen)
4. User Changes Selection
 - ↳ Repeat Step 3



Implementation Checklist



Already Done

- ☒ Landing page with routing
- ☒ Features page
- ☒ TryNow page structure
- ☒ Header and Footer components
- ☒ SearchBar component (basic)
- ☒ Material UI theme setup
- ☒ Montserrat font globally applied
- ☒ React Router setup



To Implement

- ☐ API service layer (axios integration)
- ☐ Country/Province selection dropdowns
- ☐ Analytics dashboard layout (75/25 split)

- ☐ Property Type Distribution Chart
- ☐ Room Type Distribution Chart
- ☐ Amenities Chart (top 20)
- ☐ Price Statistics Cards
- ☐ Review Statistics Cards
- ☐ Instant Booking Chart
- ☐ Listings List/Table component
- ☐ Listing Card component
- ☐ Loading states
- ☐ Error handling
- ☐ Responsive design for mobile/tablet



Future Enhancements

- ☐ Listing detail modal/page
- ☐ Advanced filtering
- ☐ Comparison mode
- ☐ Export functionality
- ☐ User preferences
- ☐ Real-time updates
- ☐ Client-side search/filter



Design Considerations

- **Layout:** 75% analytics (left/top) + 25% listings (right/bottom)
- **Charts:** Material UI Chart components (or Recharts/Chart.js with MUI styling)
- **Listings:** Scrollable with virtual scrolling for performance (if > 1000 items)
- **Responsive:** Stack vertically on mobile devices
- **Loading:** Skeleton loaders while fetching data
- **Errors:** User-friendly error messages with retry options